

Market & Demand

Market Structure

The SRT market is a differentiated oligopoly with 5 firms. Products are substitutes but not identical -- they differ in efficacy, safety, convenience, and brand reputation. Patients (and their physicians) choose among available products based on these attributes and price.

Demand System

Demand is modeled as a multinomial logit system, a standard model of differentiated product competition used in industrial organization economics.

Each quarter, a population of potential patients makes a choice:

- Buy from Firm 1, Firm 2, ... Firm 5, or
- Buy nothing (the "outside option").

The probability that a patient chooses Firm i depends on:

$$\text{Utility}_i = a * \text{quality}_i - b * \text{price}_i + g * \text{brand}_i + \text{xi}_i + \text{eps}_i$$

Where:

- quality_i = safety-adjusted composite quality score (see below)
- price_i = annual treatment price
- brand_i = accumulated brand capital (from marketing and track record)
- xi_i = firm-specific taste shock (random, drawn each quarter; std dev = 0.05)
- eps_i = individual patient taste shock (Type I extreme value -- creates the logit)

The outside option utility is:

$$\text{Utility}_0 = V_0(t) + \text{macro_shock} + \text{eps}_0$$

Where $V_0(t)$ captures the baseline reluctance to adopt (starts at 3.5, decays by 0.03/quarter as awareness grows, floor at 0.5) and macro_shock represents aggregate demand shocks (std dev = 0.08).

Quality Score and Safety Adjustment

The quality score entering the utility function is:

$$\begin{aligned} \text{raw_quality} &= w_{\text{eff}}(t) * \text{efficacy_index} + w_{\text{saf}}(t) * \text{safety_index} + w_{\text{con}}(t) * \text{convenience_index} \\ \text{effective_quality} &= \text{raw_quality} * \text{ae_demand_modifier}(\text{serious_ae_rate}) \end{aligned}$$

Where the $\text{ae_demand_modifier}$ scales demand based on the firm's serious adverse event rate -- a critical multiplier that captures the public's fear of side effects:

Serious AE Rate	Modifier	Interpretation
> 5%	0.5x	"Experimental, risky"
3-5%	0.7x	"Promising but dangerous"
1-3%	1.0x	"Acceptable" (baseline)
0.5-1%	1.3x	"Safe enough for broad adoption"
< 0.5%	2.0x	"Routine medical procedure"
< 0.1%	3.0x	"Over-the-counter safe"

The quality weights shift over time (early market values efficacy most; mature market values convenience and safety most):

Period	w_eff	w_saf	w_con
Years 1-3	0.50	0.30	0.20
Years 4-7	0.35	0.40	0.25
Years 8-12	0.25	0.40	0.35
Years 13-20	0.20	0.35	0.45

See doc 09 (Parameters and Calibration) for all numeric parameters including the demand coefficients ($a=1.0$, $b=0.000015$, $g=0.4$) and stock accumulation rates.

What This Means Intuitively

- Lower prices attract more patients, but with diminishing returns (you can't capture 100% of the market with low prices alone).
- Higher quality attracts more patients, and quality differences matter more than price differences for wealthy early adopters.
- Brand matters: patients are nervous about a new therapy. A firm with a strong brand (built through marketing and a good safety track record) is preferred over an unknown competitor, all else equal.
- Random shocks prevent deterministic outcomes: even with identical products and prices, market shares fluctuate quarter to quarter.

Market Size Over Time

The total addressable market grows as the product improves and awareness spreads.

Potential Patient Pool (global, adults 50+)

Year	Awareness Rate	Willing if Affordable	Implied Addressable
2031 (start)	15%	5% of aware	~4,500,000
2033	35%	12% of aware	~25,200,000
2036	60%	25% of aware	~90,000,000
2040	80%	45% of aware	~216,000,000
2045	95%	70% of aware	~399,000,000
2050	98%	85% of aware	~499,800,000

But affordability is the binding constraint. The "addressable" market is only the subset that can and will pay the prevailing price.

Demand as a Function of Price (Gen 1 quality, Year 1)

Price Point (annual)	Patients Who Would Buy	Revenue Potential
\$200,000	~40,000	\$8.0B
\$150,000	~75,000	\$11.3B
\$100,000	~180,000	\$18.0B
\$80,000	~300,000	\$24.0B
\$50,000	~800,000	\$40.0B
\$30,000	~2,500,000	\$75.0B
\$10,000	~15,000,000	\$150.0B

\$5,000	~40,000,000	\$200.0B
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The revenue-maximizing price depends on your cost structure and competitors' prices. At Gen 1 COGS of \$13,600/course, pricing below ~\$30,000 sacrifices margin. But a firm with Gen 3 COGS of \$2,500 can profitably serve at \$10,000 and capture enormous volume.

Market Growth Drivers

- Product improvement (lower side effects -> more people willing to try)
- Price reduction (lower COGS -> lower prices -> larger addressable market)
- Awareness (grows naturally over time + accelerated by marketing)
- Reimbursement (government/insurance coverage -- uncertain, may arrive mid-sim)
- Demographic aging (the 50+ population grows ~1.5% annually)

Market Contraction Risks

- Safety scandal (paralysis/death event -> industry-wide demand drop 20–40%

for 2–4 quarters)

- Regulatory tightening (new requirements -> reduced supply or suspended sales)
- Recession (macro shock -> discretionary healthcare spending cut 10–25%)
- Competitor substitute (non-SRT longevity approach, e.g., caloric restriction mimetic, captures some demand)

Geographic Segments

The global market breaks into distinct regions with different characteristics:

Region	Share of Addressable	Price Sensitivity	Regulatory	Growth Rate
North America	35%	Low (premium market)	FDA (ALT pathway)	Moderate
Europe	25%	Medium	EMA (slower approval)	Moderate
East Asia (Japan, Ko	25%	Medium-High	Variable by country	High
Middle East & Gulf	8%	Very Low (ultra-prem	Fast-track	Moderate
Rest of World	7%	Very High	Variable	Low initially

In the simulation, geographic segments are abstracted -- firms compete in a single global market, but demand shocks may reflect regional events (e.g., "China approves reimbursement" -> demand surge; "EU safety review" -> regional demand drop).

Brand Capital and Marketing

How Brand Works

Brand capital accumulates over time based on:

- Marketing spend (SGA): Direct-to-consumer advertising, physician education, medical conference presence.
- Track record: Quarters of operation without serious safety incidents.
- Word of mouth: Satisfied patients generate organic demand (modeled as a small positive feedback on quality * patient base).

Brand depreciates at ~10% per quarter if a firm stops marketing. It can also crash if the firm experiences a publicized safety event.

Marketing Effectiveness

Quarterly SGA Spend	Brand Capital Gain	Notes
\$0	-10% of current brand	Brand decays without maintenance
\$5M	Maintenance (0% net)	Holds current position
\$10M	+5% brand growth	Moderate investment
\$25M	+10% brand growth	Aggressive campaign
\$50M+	+12% brand growth	Diminishing returns above \$50M

Marketing is more effective when product quality is higher. Advertising a product with 7% serious AE rate is less persuasive than advertising one with 2% AE rate.

Physician Influence

In the early market (2031–2036), ~70% of treatment decisions are physician-influenced. Physician education (part of SGA) is therefore more important than consumer advertising early on. As the market matures and patient awareness grows, direct-to-consumer marketing becomes relatively more important.

Price Competition Dynamics

Price Elasticity

The market is relatively inelastic at high prices (wealthy early adopters are not price-sensitive) but highly elastic in the mid-range (the "affordability cliff" where millions of patients enter or exit based on small price changes).

Rough price elasticities by segment:

- Ultra-wealthy: -0.3 (very inelastic)
- High-net-worth: -0.8 (moderately inelastic)
- Affluent: -1.5 (elastic)
- Upper-middle: -2.5 (very elastic)
- Mass market: -4.0+ (hyper-elastic)

Strategic Implications

- Early years (Gen 1): Price high (\$80K–\$150K), serve the wealthy, maximize margin per patient. Volume is constrained by capacity anyway.
- Mid-years (Gen 2): Lower prices to \$30K–\$60K as COGS drops, capture the affluent segment. Volume growth drives revenue even as margin per unit declines.
- Late years (Gen 3+): Price at \$5K–\$15K, serve the mass market. Volume is enormous. The winner is the firm with the lowest COGS and broadest distribution.
- Undercutting is risky: Cutting price below COGS to gain share is suicidal.

The financial market will punish unprofitable pricing with lower equity valuations and tighter credit.

Demand Shocks (Environment-Generated)

Each quarter, the environment draws random shocks that affect demand:

Macro Demand Shock (affects all firms equally)

- Mean: 0, Std Dev: 0.08
- Represents: economic conditions, media coverage, regulatory mood
- Range: roughly -20% to +20% in extreme quarters

Firm-Specific Taste Shock (ξ_i)

- Mean: 0, Std Dev: 0.05
- Represents: word of mouth, physician preference shifts, regional variation
- Independent across firms

Safety Event Shock (rare but large)

- Probability: ~3% per firm per quarter of a "publicized adverse event"
- Effect: -15% to -40% demand for that firm for 2–4 quarters
- Industry spillover: other firms also lose 5–10% demand (guilt by association)

These shocks ensure that outcomes are not purely deterministic, even when firms make similar decisions.

What Firms Should Consider

- You are creating the market, not just competing in it. Early investment in R&D and marketing grows the total pie, benefiting everyone -- but benefiting you most if you lead.
- Price is a strategic choice, not just a margin calculation. Pricing below competitors signals aggression and may trigger a price war. Pricing above signals premium quality but risks losing volume.
- Demand is noisy. Don't overreact to a single bad quarter. Look at trends over 4–8 quarters before concluding your strategy isn't working.
- The outside option is your biggest competitor initially. Most potential patients are choosing "do nothing" -- not choosing a rival firm. Your marketing and pricing should focus on converting non-adopters, not just stealing share from competitors.
- Market share is sticky but not permanent. Patients who start on one firm's product tend to stay (switching costs are real -- new titration, monitoring). But a clearly superior product will overcome switching costs within 2–3 quarters.